Circuit model and solder practice

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Learning target: Test a circuit representation before making permanent connections.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 17.*
Check circuit/secured parts/clear wheels with teacher approval. Measure distance and elapsed time or another planned outcome.

02. Worked example: explain the model
One loop through the motor; switch interrupts it; no direct low-resistance bypass across battery.

03. Read and interpret the model
Closed loop permits current; open switch interrupts the conducting path.

04. Apply the physics
0.60 W; simulation observation must come from the student's trial.

05. Explain your reasoning
Student records actual feedback, such as secure connection, wetting or excess solder; no invented approval.

06. Review: connect an earlier idea *Review: Unit 7, Day 8.*
1 A, by charge conservation.

07. Exit check
Disconnected; avoid unintended current, shorts and powered motion during work.

08. STRETCH: test the model
Open path gives no steady current; bypass risks excessive current. Disconnect power, inspect continuity/routing under classroom procedure.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 17.*
20 J may go to wheel rotation, heating, friction, drag or sound. Total energy accounting includes these terms.

02. Worked example: explain the model
One loop through the motor; switch interrupts it; no direct low-resistance bypass across battery.

03. Read and interpret the model
Closed loop permits current; open switch interrupts the conducting path.

04. Apply the physics
20 Ω; 1.8 W; motor rotation generates back emf and behavior depends on load and speed.

05. Explain your reasoning
Student records actual feedback, such as secure connection, wetting or excess solder; no invented approval.

06. Review: connect an earlier idea *Review: Unit 7, Day 8.*
Parallel 6 Ω; series 24 Ω.

07. Exit check
Appearance alone does not establish wetting, continuity or mechanical strength; use approved inspection/testing.

08. STRETCH: test the model
Running 2 A, 12 W; stalled 6 A, 36 W. Effective motor behavior includes back emf; do not intentionally sustain a stall.